NAME: _____ Geometry (FL B.E.S.T.)

4.1 Classifying Triangles & Angle Sum

Triangles are classified two independent ways. **By sides:** scalene (no two sides congruent), isosceles (at least two sides congruent), equilateral (all three sides congruent). **By angles:** acute (all three angles <90^\circ), right (exactly one $90°$ angle), obtuse (exactly one angle >90^\circ), equiangular (all three angles congruent, which forces equilateral). The **Triangle Angle-Sum Theorem** guarantees the interior angles of every triangle total $180°$; its corollary, the **Exterior Angle Theorem**, says an exterior angle equals the sum of the two non-adjacent (remote) interior angles. Because the angle sum is fixed at $180°$, a triangle can never have two right angles or two obtuse angles, and the acute angles of a right triangle must be complementary. This worksheet blends definitions, algebra, coordinate geometry, and short proofs.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. A triangle has sides $9, 9, 9$. Classify it by sides.

2. A triangle has sides $6, 9, 11$. Classify it by sides.

3. A triangle has angles $100°, 40°, 40°$. Classify by angles.

4. A triangle has angles $90°, 45°, 45°$. Classify by angles.

5. A triangle has angles $65°$, $75°$, and x. Find x.

6. The angles of a triangle are in ratio $1 : 2 : 3$. Find each angle and classify by angles.

7. In $\triangle ABC$, $\angle A = 2x + 5$, $\angle B = 3x - 1$, $\angle C = 91°$. Find x, each angle, and classify by angles.

8. The two remote interior angles are $48°$ and $62°$. Find the exterior angle.

9. An exterior angle is $125°$; one remote interior angle is $70°$. Find the other remote angle.

10. Exterior angle $= 4x$; remote angles are $2x + 10$ and $x + 20$. Find x and the exterior angle.

11. Vertices $A(0, 0)$, $B(6, 0)$, $C(0, 8)$. Find all three side lengths and classify by sides and angles.

12. Explain why a triangle cannot have two right angles.

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13. In isosceles $\triangle ABC$ with $AB = AC$, the exterior angle at B measures $150°$. Find $\angle B$, $\angle C$, and $\angle A$, then classify by angles.

14. Vertices $A(1, 1)$, $B(5, 1)$, $C(1, 5)$. Find all side lengths (exact form) and fully classify the triangle by sides and angles.

15. In $\triangle ABC$, $\angle A = \angle B$ (isosceles), and the exterior angle at C measures $100°$. Find $\angle C$, then $\angle A$ and $\angle B$, and classify by angles.

✓ Answer Key

1. Equilateral
2. Scalene
3. Obtuse
4. Right
5. $x = 40°$
6. $30°, 60°, 90°$ — right triangle
7. $x = 17$; angles $39°, 50°, 91°$ — obtuse triangle
8. $110°$
9. $55°$
10. $x = 30$; exterior angle $= 120°$
11. $6, 8, 10$; right scalene triangle
12. Two $90°$ angles already total $180°$, leaving no positive measure for the third angle — contradicting the Angle-Sum Theorem
13. $\angle B = \angle C = 30°$, $\angle A = 120°$ — obtuse isosceles
14. $AB = AC = 4$, $BC = 4\sqrt{2}$; isosceles right triangle (right angle at A)
15. $\angle C = 80°$, $\angle A = \angle B = 50°$ — acute isosceles